

Improved Photoplethysmography-Based Four-Stage Sleep Classification with Explainable AI-Driven Machine Learning

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Abstract—Single-sensor-based multistage sleep classification is a challenging task to acquire the accuracy of traditional polysomnography (PSG). Feature engineering is the key factor in improving the accuracy of any measurement based on machine learning with a single sensor. However, too many features increase the complexity of the model and training time. To address this challenge, explainable Artificial Intelligence (XAI) is used in several machine learning classifiers to determine the most explainable features of single-sensor-based (PPG) data for a lightweight model. This study demonstrated substantial improvements in prediction accuracy 75.54% for 4-stage sleep classification utilizing 42 PPG data variables. With a simpler AI model and a single low-cost PPG sensor, our adaptive approach shows promise for accurately staging sleep. This may simplify the detection and treatment of sleep disorders, potentially allowing for at-home solutions.

Index Terms—Photoplethysmography, Single sensor, Machine Learning Classifiers, Explainable AI, Home Sleep Analysis, Sleep Staging

I. INTRODUCTION

Multiple stages of sleep are necessary for both physical and cognitive recovery. Mostly, sleep is classified into wake, non-REM sleep and rapid eye movement (REM) sleep, where non-REM sleep can be further divided into light and deep sleep [1]– [4]. These stages are critical for memory consolidation, hormone regulation, and immune support [5]– [6]. Even a neurological condition may lead to obstructive sleep apnea (OSA) [7]– [8]. The focus of this study is to examine these issues in individuals who has sleep apnea.

Using polysomnography (PSG), which records brain waves (EEG), eye movements (EOG), muscle contractions (EMG), heart rate, breathing rate, and depth during an overnight stay in a sleep lab, sleep staging is the process of classifying an individual's sleep cycle throughout the night [9]– [10]. Manual data collection and classification into 30-second

epochs is done in accordance with guidelines provided by the American Academy of Sleep Medicine (AASM). Non-REM (NREM) sleep, often known as the wake stage, is divided into three categories based on the data gathered: NREM stages 1, 2, and 3 [11]. Long-term monitoring is difficult as data can be gathered manually by a clinician. The overall process is time-consuming and restricted to a single night stay at the clinic [12]. Roughly 93.6% of sleep centers stopped doing in-person testing during the COVID-19 outbreak and shifted to telemedicine [13].

The study proposes a single photoplethysmographic (PPG) sensor-based sleep stage categorization approach as a more practical, at-home alternative to current multi-sensor methods. PPG measures variations in blood flow in tissues and offers important information on breathing patterns, cardiovascular dynamics, and heart rate variability during sleep stages. Continuous PPG monitoring is now achievable because to recent advances in wearable technology, paving the way for more scalable sleep research. However, PPG lacks direct measures of brain activity and muscle tone, which are essential for accurately differentiating between REM and NREM sleep stages [14]. Therefore, excellent feature extraction from PPG data is crucial to match conventional polysomnography (PSG) accuracy. In this paper, initially we investigated multiple machine learning classifiers using 72 PPG characteristics to meet the difficulty [15].

Over the last decade, machine learning methods have been often utilized to categorize sleep stages based on PPG signals. Some common classifiers, such as Support Vector Machines (SVM), has shown useful in managing variables in SVM associations that can be defined as complex and nonlinear. Other models, such as the ensemble approach Random Forest (RF), provide excellent prospects to comprehend various

stages of sleep classification due to their ability to handle vast amounts of data and complicated features. Gradient boosting techniques like eXtreme Gradient Boosting (XGBoost) and Categorical Boosting (CatBoost) are gaining prominence due to their superior performance in this area.

Using classifiers such as RF, K-Nearest Neighbors, and SVM on a PPG sensor resulted in 72.23% accuracy for four-stage sleep detection [15]. A RF with 285 participants obtained 98% accuracy [16]. Ucar et al. used PPG alone to create an SVM classifier that accurately differentiated between sleep and awake stages 77% accuracy [17]. Huttanen et al. used a similar technique on a PPG-based sleep staging algorithm and achieved an accuracy of 83.3% on 3-class, 74.1% on 4-class, and 68.7% on 5-class models [18].

Gradient Boosting algorithms like as XGBoost and CatBoost have recently gained popularity for their improved performance in a variety of classification problems. XGBoost using only three features from PPG and activity signals achieved an F-1 score of 83.88% for 2-stage sleep classification [19]. XGBoost, in particular, has been widely embraced due to its effectiveness, scalability, and ability to handle missing data. However, because CatBoost is primarily built to work with categorical variables, it is an effective tool for dealing with mixed-type data, which is commonly present in PPG datasets.

To obtain the same level of accuracy as PSG-based approaches for four-stage sleep classification, the PPG-based approach requires significant feature engineering and statistical analysis. Initially, 84 features were retrieved from raw PPG data using feature engineering [15]. After statistical analysis, 72 features are used to machine learning classifiers such as SVM, RF, CatBoost and XGboost.

This study outperformed prior studies [15] with an accuracy of 75.29% for CatBoost, 75.15% for XGBoost, 74.41% for RF, and 70.20% for SVM in 4-stage sleep categorization.

CatBoost exceeds the other machine learning model in terms of reliably and accurately classifying the four stages of sleep. However, rigorous statistical methods are difficult and time-consuming to determine the largest amount of relevant features that match the sleep staging accuracy of a single sensor-based measurement. Another disadvantage is that having too many features in a model makes it challenging and lacks explainability. Addressing these issues, explainable Artificial Intelligence (XAI) is applied to select the most explainable features while reducing model complexity. Another solution is to use Logistic Regression, but unlike XAI it can not capture and explain complex, non-linear patterns that often exist in PPG data. XAI aids in ensuring fairness, detecting bias, and validating feature contributions, which is especially crucial in high-stakes industries such as healthcare. In physiological signal analysis, XAI is gaining traction, particularly for understanding healthcare data in wearable sensors, allowing both practitioners and patients to better comprehend the importance of various physiological signals. Studies show that XAI is becoming increasingly important in fields such as engineering, consumer products, and

healthcare. Researchers use XAI approaches to improve model interpretability while preserving high prediction accuracy in popular algorithms such as RF, XGBoost, and SVM [20]-[22]. Explainable AI (XAI) in physiological signal analysis prioritizes explanatory models above interpretable ones [23]. XAI aims to bridge the gap between machine learning models and biologically/clinically interpretable data in biomedical signal processing [24]. XAI is particularly important for understanding machine learning classifiers such as CatBoost, XGBoost, SVM, and RF, which are frequently used as black boxes [20]. To overcome this issue, XAI is applied to each of the classifiers to find the most essential features while maintaining accuracy. XAI on these models includes built-in feature importance methods that reveal how specific features influence model predictions, boosting interpretability and trust.

The CatBoost classifier model achieves the highest accuracy of 75.54% with only 42 features after XAI.

The explainable AI in our machine learning model indicates that improved accuracy and explainability can be achieved with reduced features without requiring in-depth feature engineering and statistical analysis.

The proposed study includes the following contributions:

- Statistical analysis to reduce the feature counts.
- Comprehensive evaluation of SVM, XGBoost, CatBoost, and RF classifiers on PPG data to identify four sleep stages, incorporating reduced and explainable features generated from explainable AI for PPG-based wearable devices.

The remainder of the paper is structured as follows: The Methodology section provides information about the dataset, process flow, and model development. The results show model performance measures, and the discussion analyzes the findings, limitations, and implications. The conclusion summarizes the findings and suggests future research directions.

II. METHODOLOGY

A. Dataset

This study used PPG data from ten individuals recorded for a single night (7-10 hours) and PSGs taken from patients with sleep-disordered breathing at Charite Hospital, Berlin, Germany [25]. The PSG recordings are separated into 30-second epochs and classified to six stages according to Rechtschaffen and Kales (R&K) method.

R&K categorizes sleep into six stages: Wake, REM, Light Sleep (S1 and S2) and Deep Sleep (S3 and S4). Our study aims to identify four stages of sleep using standard R & K scoring. Thus, S1 and S2 are classed as Light Sleep (LS), whereas S3 and S4 are labeled as Deep Sleep (DS). Therefore, the light and deep sleep stages can be compared to the Standard AASM scoring system, which classifies deep sleep as a single stage (N3) and light sleep as N1 and N2. Fig. 1 shows the segmentation of annotated PPG data by Apnea-Hypopnea Index (AHI) score.

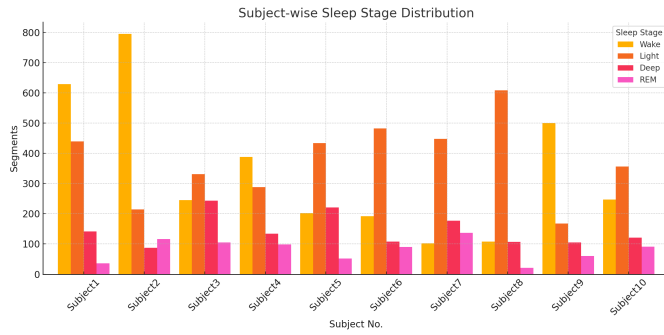


Fig. 1. Dataset distribution.

B. Data Pre-processing

The PPG data, initially recorded at 128 Hz, is downsampled to 64 Hz to improve calculation speed and reduce model complexity, focusing on capturing overall sleep patterns rather than fine-grained features [26]. The dataset consists of 9,428 sleep epochs, each with a 30-second period. After detrending, Z-score is used to standardize segment lengths ranging from 815 to 1,246, along with 84 features [25]. Statistical analysis was used to reduce the feature count to 72, removing features with similar values distributed throughout the column.

C. Process flow: Machine Learning approach

The data is then split into training and testing where 70% of the data allocated for training and 30% for testing. Out of 9428 samples, 6660 are utilized for training and 2828 for testing.

1) *Model Training*: The segmented data undergoes various machine learning classifiers, such as SVM, RF, CatBoost, and XGBoost to identify the optimal model. Among these, CatBoost achieved the highest accuracy, owing to its robust handling of categorical features and boosting approach. Each model is configured with specific hyperparameters optimized for performance, based on prior experimentation. These configurations include parameters such as the number of estimators, learning rates, maximum depth for tree-based models, and kernel-related parameters for SVM. The accuracies are obtained with using 72 features. The best model is selected based on its performance in classifying the data into four sleep stages: Wake, LS, DS, and REM. Fig. 2 depicts the overall process flow diagram for the data training and testing. Table I shows the accuracy across all models with only hyperparameter tuning.

2) *Model Training and Explainable AI (XAI)*: Several XAI techniques were used in this phase to train the models with its most important features.

Built-in feature Importance on Random Forest, XGBoost, and CatBoost

Tree-based models quantify each feature's decision-making value automatically. Reduced entropy (information gain) or Gini impurity in Random Forest and XGBoost prioritize features. In boosting iterations, CatBoost estimates feature contribution significance values. In this study, each model was

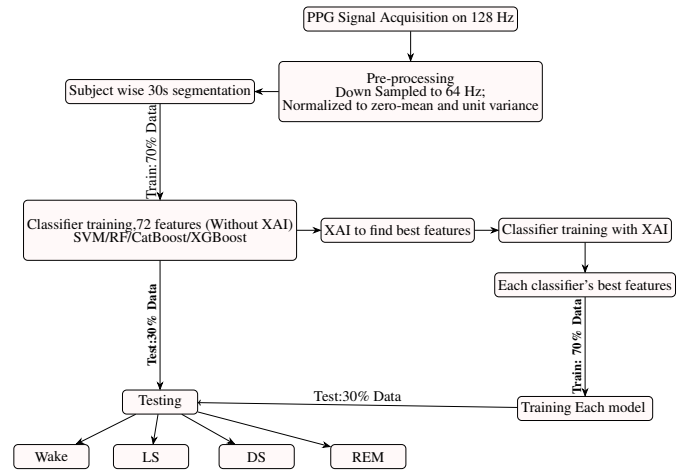


Fig. 2. Process flow for 4-stage sleep classification.

trained and tested five times using a 70-30 train-test split with random states to introduce data-split variability for reliability. After each repetition, significance scores are averaged in all runs, stabilizing features.

Permutation Importance on SVM

Permutation importance is used for feature scoring in SVM. This approach randomly shuffling feature values and measuring accuracy changes assesses each feature's influence on model performance. Features that greatly degrade model accuracy when permuted are crucial. To ensure robustness, permutation significance is calculated on the test set 10 times. Five random splits of the data are averaged to obtain the estimation of consistent feature significance.

3) *Iterative Assessment of Models and Feature Importance Ranking*: Every model uses iterative feature selection to improve prediction. The appropriate amount of features for model accuracy is determined by testing model performance across multiple subsets of features, ordered by relevance. For each model, the features are ordered in descending order by significance score to aid in feature selection and testing.

4) *Iterative Evaluation Across Feature Subsets*: Each model is iteratively trained and tested using progressively larger subsets of the sorted features. The number of features used in each iteration ranges from the single most important feature to the complete set of features. In each iteration, the following steps are carried out:

Feature Subset Selection: A subset of the sorted features, beginning with the top-ranked feature and incrementally increasing the number of features, is selected for training and testing.

Data Splitting: The dataset is split into training and testing sets using a 70-30 ratio, with a fixed random state to ensure the consistency and reproducibility of results across iterations.

Model Training and Evaluation: The model is trained on the selected subset of features and evaluated on the test set. The evaluation metric used is accuracy, which measures the model's ability to correctly classify the sleep stages based on the given features.

Accuracy Recording: The accuracy obtained for each subset of features is recorded, and this process is repeated for all feature subsets to identify performance trends.

D. Model Evaluation and Analysis

All XAI best feature models are assessed using accuracy, precision, recall, F1-score, confusion matrices and ROC curves. Metrics performance for each sleep stage are assessed in classification reports. XAI identifies key model prediction properties. Performance optimisation enhances interpretability and model resilience for real-world applications. Due to rigorous testing, models trained on XAI-selected features accurately and effectively classify sleep stages. Model performances are also assessed using Cohen's Kappa (κ), which is needed to measure the agreement between predicted and actual labels. Equation [1-3] is used in this study to calculate Cohen's Kappa.

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (1)$$

$$p_o = \text{the proportion of correctly identified test cases to all test cases} \quad (2)$$

$$p_e = \sum_{i=1}^k \frac{(TP_i + FP_i) \cdot (TP_i + FN_i)}{n^2} \quad (3)$$

III. RESULT

The performance of four different machine learning models such as RF, SVM, XGBoost, and CatBoost is assessed thoroughly with 72 features and hyperparameter tuning for multi-class classification. All these models are then tested with XAI for four stage classification.

- 1) **RF:** Recorded 74.41% accuracy with 72 features with precision (0.70-0.80), recall (0.40-0.84), and F1-score (0.53-0.81). After applying XAI, accuracy is increased to 75.82% with only 51 features. There is also a small improvement in precision (0.83-0.72), recall (0.45-0.84) and F1-score (0.58-0.81).
- 2) **SVM:** Lowest accuracy at 70.20% with precision (0.58-0.78), recall (0.66-0.76), and F1-score (0.62-0.77). Accuracy improved after XAI to 71.40% with 48 features with a precision (0.59-0.80), recall (0.66-0.78), and F1-score (0.64-0.79).
- 3) **XGBoost:** Achieved better accuracy of 75.93% with an increase in precision (0.72-0.79), recall (0.45-0.85), and F1-score (0.57-0.82) with 56 features. Without XAI, accuracy was 75.15% with precision (0.67-0.77), recall (0.66-0.70), and F1-score (0.57-0.82). Excelled in classifying REM.
- 4) **CatBoost:** Achieved 75.29% accuracy, precision (0.73-0.83), recall (0.50-0.86), and F1-score (0.62-0.81). After XAI, CatBoost depicts 75.54% accuracy, precision (0.72-0.79), recall (0.51-0.83) and F1-score (0.62-0.81) with 56 features. Particularly effective in the REM.

Table I shows the summary of accuracy, precision, recall and F1-score of each model with 72 features and Table II depicts

TABLE I. PERFORMANCE METRICS OF DIFFERENT MODELS WITH HYPERPARAMETER TUNING.

Model	No. of Features	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
RF	72	74.41	77	66	69
SVM	72	70.20	67	70	68
XGBoost	72	75.15	76	67	70
CatBoost	72	75.29	77	69	72

TABLE II. PERFORMANCE METRICS OF MODELS WITH XAI.

Model	No. of Features	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
RF	51	75.82	76	76	75
SVM	48	71.40	72	71	72
XGBoost	56	75.93	76	76	76
CatBoost	42	75.54	76	76	75

all the assessment values of each classifier after applying XAI with best features. After comparing between these two tables, it is certain that all models provide better accuracy, precision, recall, and F-1 score for a reduced number of features.

A. Confusion Matrix

The confusion matrix analysis for RF, SVM, XGBoost and CatBoost models indicates variations in classification accuracy across sleep stages. Among the models evaluated, XGBoost has the highest accuracy of 75.93%, followed by RF 75.82% and CatBoost 75.54%, with SVM lagging at 71.40% in Fig. 3. All models perform well in classifying the Wake and LS stages, but XGBoost and RF, in particular, misclassify these two stages significantly. Moderate confusion exists between the REM and DS stages, with models frequently misclassifying them as Wake and LS. In comparison to ensemble approaches, SVM has a higher rate of misclassification, particularly between Wake and REM. Overall, XGBoost outperforms the other models, with a minor advantage in addressing stage confusion.

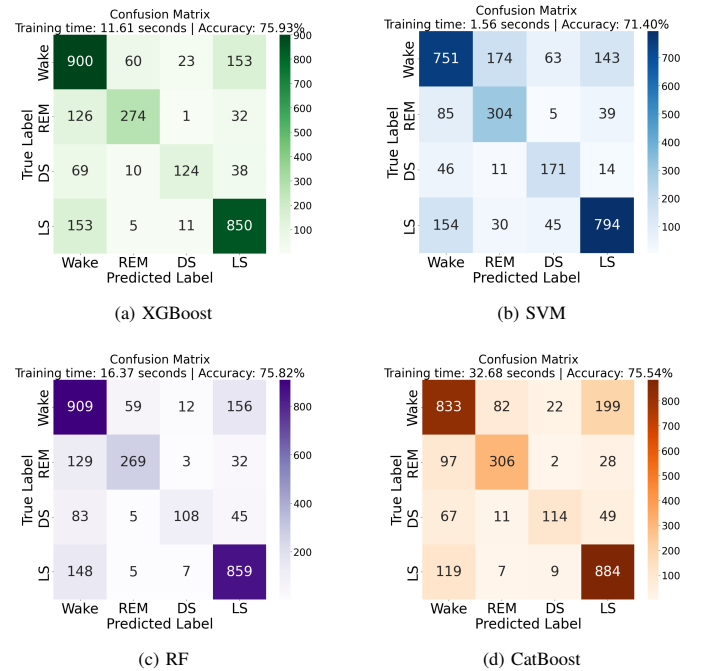


Fig. 3. Comparison of Confusion Matrices for XGBoost, SVM, RF, and CatBoost.

B. ROC Curve Comparison

The receiver operating characteristic (ROC) curve in Fig. 4 show that all models perform well in classification, with RF, XGBoost, and CatBoost obtaining AUCs of 0.93 and SVM following in close at 0.92. Despite its high performance, RF deviates slightly from the ideal curve, indicating some false positives. SVM performs well, however it is less precise in class discrimination than ensemble models. XGBoost and CatBoost show consistent performance, with low misclassification and balanced precision. Overall, all four models perform well, with slight differences in AUC values.

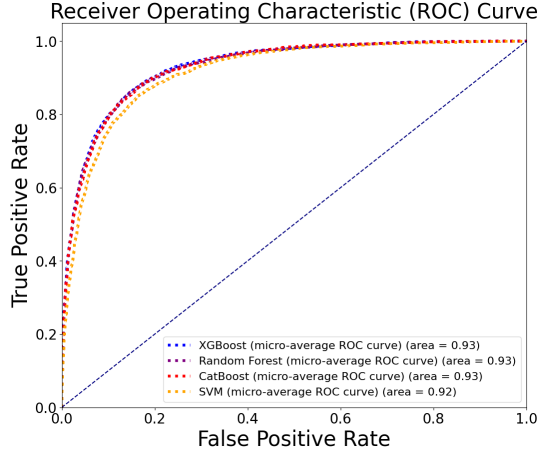


Fig. 4. ROC Curve with XAI.

IV. DISCUSSION

Table III interprets that all proposed work classifiers outperformed previous studies in terms of precision and F-1 score for a four-stage sleep classification [15], [27]. The empirical examination of hyperparameter tuning in the selected models, as well as the statistical analysis of feature selection, exceeded all previous research. In this study, explainable AI (XAI) techniques are applied to improve performance, interpretability, feature selection, and reduce computational costs in each machine learning classifier.

The proposed model exceeded previous research [15] by attaining 75.29% precision with CatBoost, 75.15% with XGBoost, 74.41% with RF, and 70.20% with the SVM classifier for 4-stage sleep categorization using 72 features. In this study, it is found that employing built-in feature importance on ensemble models led to an improved accuracy in 4-stage sleep classification: 75.54% for CatBoost with 42 features, 75.93% for XGBoost with 56 features, and 75.82% for RF with 51 features. Using permutational feature significance, SVM achieved 71.40% accuracy with 48 features. After XAI, all the proposed classifiers performed better with less features, but CatBoost consistently and reliably detected the four phases of sleep better than the other machine learning model.

One of the study's weaknesses is the limited sample size of 10 individuals with hypopnea, which may affect how well the

models generalize to newly collected data. Data augmentation could be used to improve this limitation. It is not used in this study because, for a small dataset, it may generate false patterns that do not correctly reflect the true variability of PPG data in real-world circumstances.

This study excluded deep-learning models because the dataset is small and not expanded for a single second measurement. Because transitions between sleep stages might occur in minutes [28], all data are separated into global 30-second epochs. Furthermore, when compared to deep learning techniques, using only machine learning models can make it more difficult to find complicated patterns in PPG data. Our goal is to identify the most explainable features for usage in a single pulse oximetry sensor. XAI facilitates the discovery of hidden links and the comprehension of PPG data patterns by emphasizing feature importance and explaining model decisions. Despite improvements, issues in validating XAI models persist, and future research should focus on improving evaluation methodologies while maintaining transparency, inclusion, and reliability [23]. Another limitation arising from the small dataset is the use of 42 features to achieve PSG accuracy even after utilizing explainable AI. Several other studies reported similar accuracy (77%) using only 21 features [29]. However, with additional data, XAI can better recognize unnecessary or redundant characteristics, allowing for more efficient feature selection. While XAI can not directly reduce features, increased comprehension from a larger dataset can aid in informed feature reduction. Thus, including a larger dataset will improve XAI performance.

We intended to test our methods on a bigger dataset that included both healthy adults and patients with sleep disorders. Furthermore, combining deep learning models with transfer learning from larger datasets could potentially address the issues associated with small dataset sizes by leveraging pre-trained knowledge to improve model accuracy and robustness.

V. CONCLUSION

In this study, we evaluated the impact of XAI on multiple machine learning models for four-stage sleep classification using PPG data. CatBoost outperformed state-of-the-art techniques with its accuracy of 75.54% and F1-Score of 0.75, despite only having 42 characteristics. CatBoost and other traditional machine learning models are easier to build and have lower processing costs, but their small dataset size and high feature dimensionality limit generalizability and durability. Incorporating XAI can help address this constraint by offering insights into which features are most relevant, allowing for better feature selection and reducing dimensionality, increasing the model's generalizability and resilience across datasets.

Despite these challenges, machine learning can produce promising results for single-sensor systems such as PPG by optimizing feature selection. Investigating cutting-edge methods such as transfer learning has the potential to increase model performance and better capture the subtle patterns

TABLE III. SUMMARY OF STATE OF THE ART OBTAINED RESULTS FOR THE 4 STAGE [WAKE-LIGHT-DEEP-REM] SLEEP CLASSIFICATION.

	Accuracy(%)	Precision (%)	Recall (%)	F1-Score(%)	κ	Dataset	Method
Wu et al.[2020] [27]	62	-	-	-	0.41	Private(N=31)	PPG with SpO ₂ , 55 features, SVM-ANN
Motin et al.[2023] [15]	70.19 $\pm$ 0.32	-	-	62	-	Private(N=10)	PPG, 72 features, SVM
Motin et al.[2023] [15]	72.39 $\pm$ 0.48	-	-	62	-	Private(N=10)	PPG, 72 features, RF
Motin et al.[2023] [15]	72.23 $\pm$ 0.38	-	-	65	-	Private(N=10)	PPG, 72 features, KNN
Proposed Model	75.54	76	76	76	0.59	Private(N=10)	PPG, 42 features, CatBoost with XAI
Proposed Model	75.93	76	76	76	0.60	Private(N=10)	PPG, 56 features, XGBoost with XAI
Proposed Model	75.82	76	76	75	0.60	Private(N=10)	PPG, 51 features, RF with XAI
Proposed Model	71.40	72	71	72	0.50	Private(N=10)	PPG, 48 features, SVM with XAI

present in PPG data. Improving these models may lead to more precise and effective sleep monitoring systems in practical circumstances.

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